

HAT-P-12b

R-0734

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_250213 · created 2026-07-28T01:18:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460720.0037 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.053
Transit depth (Rp/Rs)^2	3.49 +/- 1.97 [%]
Orbital Inclination (inc)	89.85 +/- 1.64
Airmass coefficient 1 (a1)	0.781 +/- 0.043
Airmass coefficient 2 (a2)	0.23 +/- 0.35
Scatter in the residuals of the lightcurve fit is	5.58 %
Best Comparison Star	#1 - [501, 286]
Optimal Aperture	2.29
Optimal Annulus	8.3
Transit Duration (day)	0.0957 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.187 ± 0.053 vs 0.1406 (deviation 0.88σ)
Duration fit vs published	0.0957 d vs 0.0975 d (deviation 0.23σ)
Mid-time O–C	1.5 min
Depth SNR	3.5
Residual scatter	5.58 %

Light curve

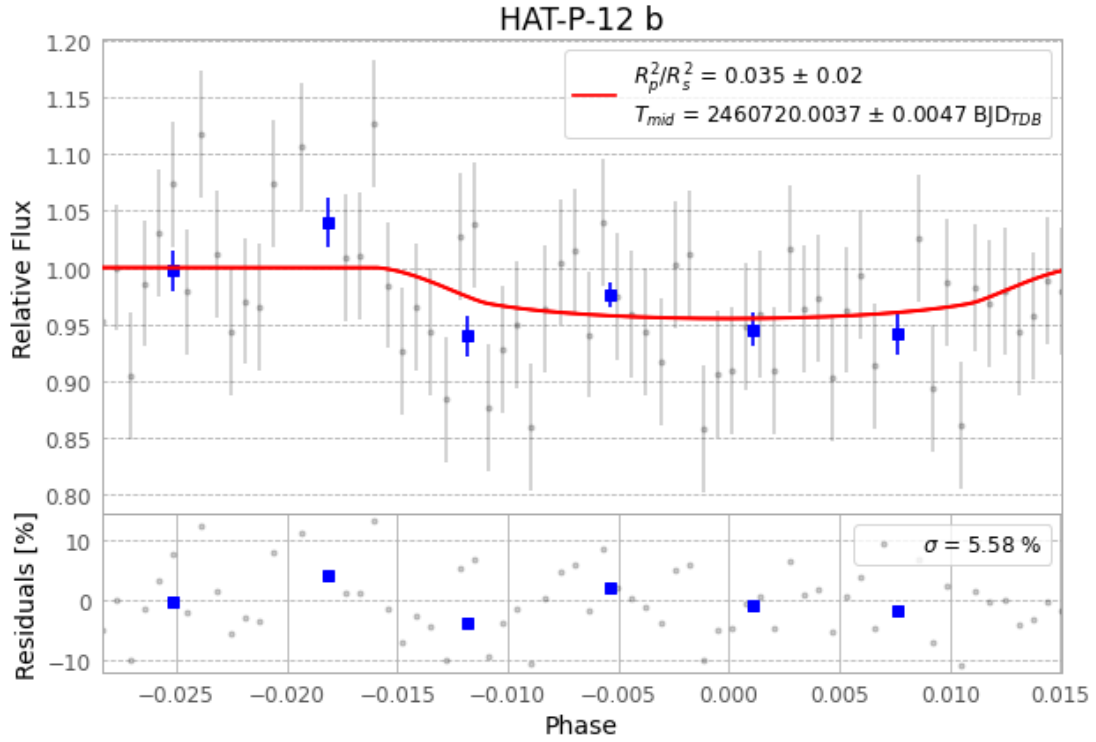


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-02-13.png (SHA-256 5a9dd99bb19f925e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a0c5a9c64b86a852...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

68 FITS frames (45 MB), HATP-12250213094818.FITS → HATP-12250213130915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250213.log (cba3359263f0...), FinalLightCurve_HAT-P-12 b_2025-02-13.png (5a9dd99bb19f...), FinalLightCurve_HAT-P-12 b_2025-02-13.csv (e56c3255ae4c...)

Compute resources

Wall time 220.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:18Z.